# Classroom Experiments

Instructor guide: setting up the class, approving designs, running the live session and collecting the data

Groups of students design randomized experiments as surveys; on class day each group presents while everyone answers its survey on their phones and the results appear on the projected dashboard. This guide covers everything on the instructor side. Students have their own guide (`docs/student-guide.pdf`).

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## 1. Creating the classroom

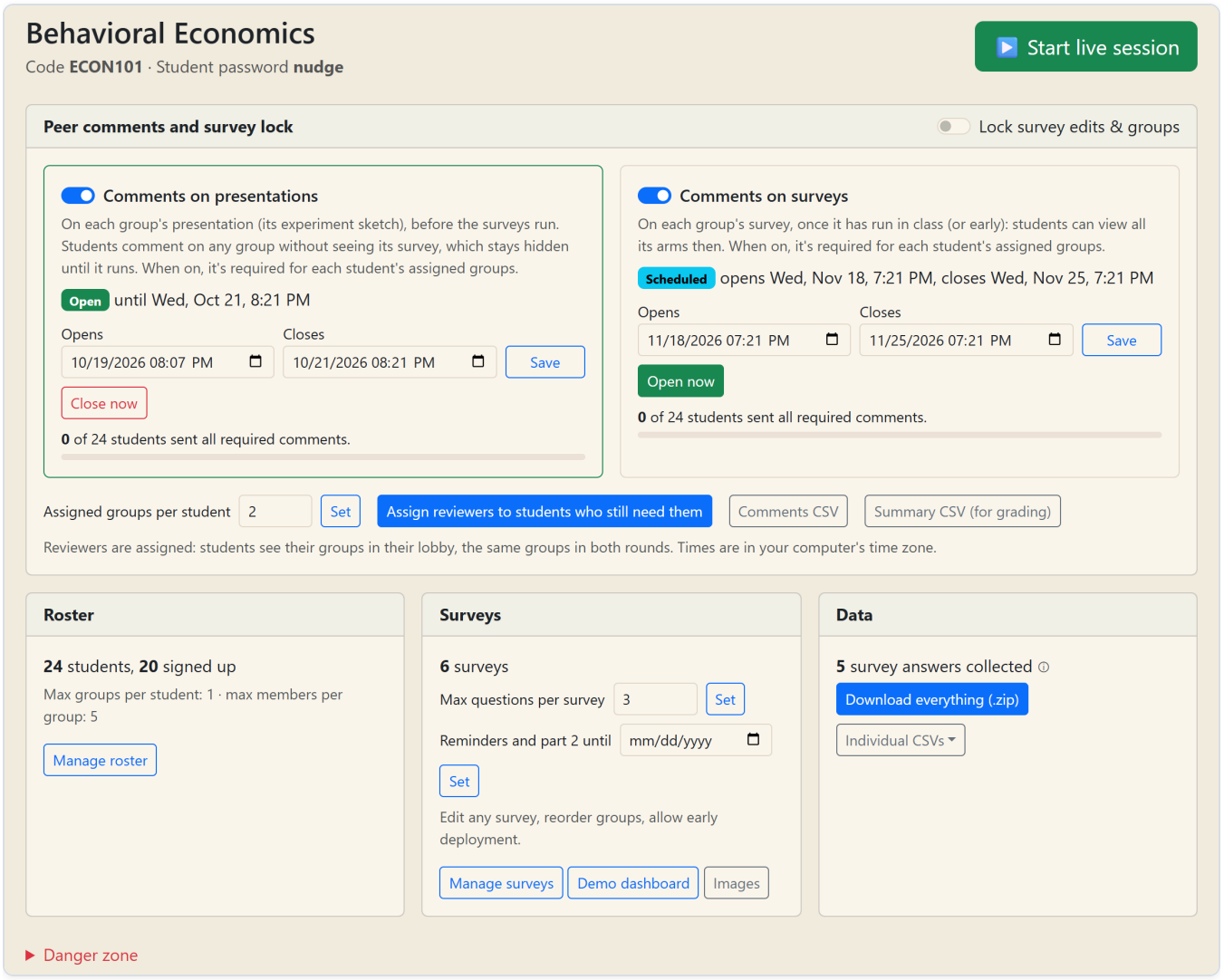
On the site's home page, choose **Create Classroom** and give it:

- a **code** students type to join (e.g. `ECON101`; unique, not case-sensitive);
- a **host password**, only for you: it opens the instructor pages;
- a **classroom password** that you share with the class (it's also in the QR code on the live dashboard);
- the **roster**: a CSV with the students' IDs and names. A Canvas gradebook export works as is.

Later, log in from **Host a Classroom** on the home page with the code and the host password. Students sign in with their SIS ID; the first time they confirm their first name and choose a password, so names and IDs in your data always match the roster.

## 2. The instructor home

Everything between classes is on the instructor home: peer comments and the survey lock, the roster, the surveys and the data. **Start live session** opens the page you project in class.

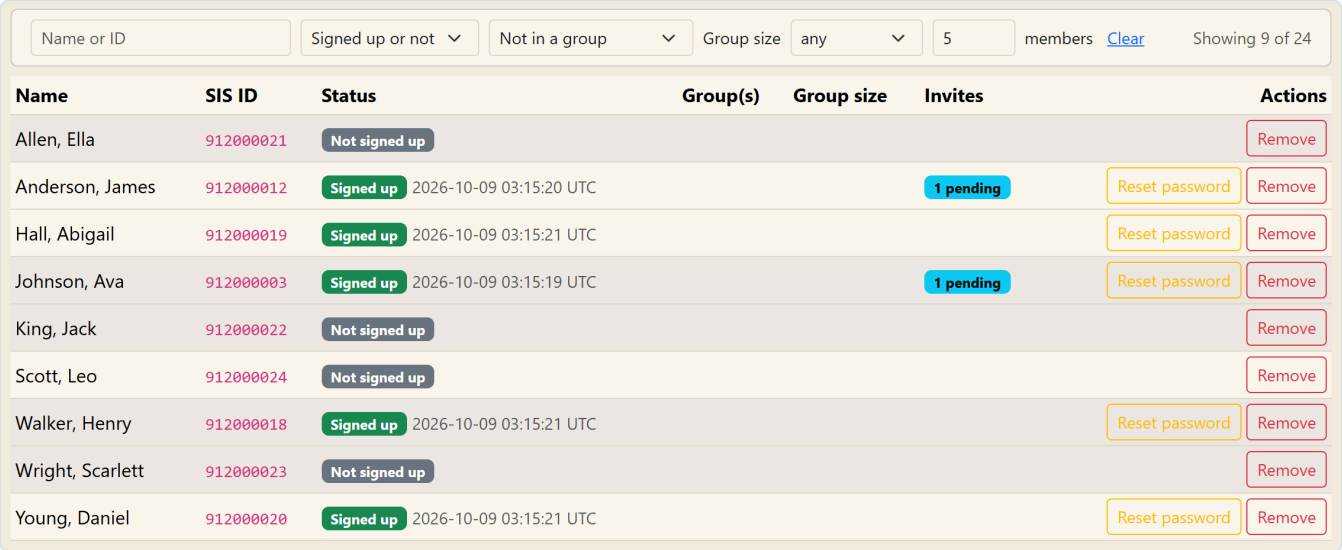



- **Max questions per survey** (Surveys card): students can't save more; you're exempt.
- **Reminders and part 2 until**: the latest date any group's notifications or remote parts can fall on. Leave it empty for no limit.
- **Two-part dates and reminders to approve** lists groups waiting for you (see section 6).
- **Survey answers collected** counts each student who answered a survey once. It stays at 0 until surveys run (in class, launched on their own, opened early or a remote part), however many surveys groups have built.

## 3. The roster

**Manage roster** shows every student, whether they signed up, their group, its size and any pending invites. The filters at the top narrow the list: name or ID, signed up or not, in a group or not, with a pending invite, alone in their group, no group and no pending invite or request (students to contact directly), asked to join a group, listed in the group finder, or group size fewer than, exactly or more than a number of members. "Showing X of Y" counts what's left. Hovering over a group number shows its members and invites; hovering over "N pending" shows who invited the student.

**Group finder.** Students without a group can opt in to a list in their lobby (with an optional note), and groups with room can list themselves as looking for members. Students can invite listed classmates, start a group with one, or ask to join a listed group (any member accepts). A small group can also propose merging into another listed group: if any member of that group accepts, the proposing group's members move over and its survey is deleted (only if everyone fits within the size limit and the proposing survey has no responses). Only those who opt in are shown; students leave the list once they join a group, get an invite or ask to join one, and a group leaves it once it's full. The roster marks "Looking for a group" and "asked to join N" (with filters), and Manage surveys marks "Looking for members" and pending requests.

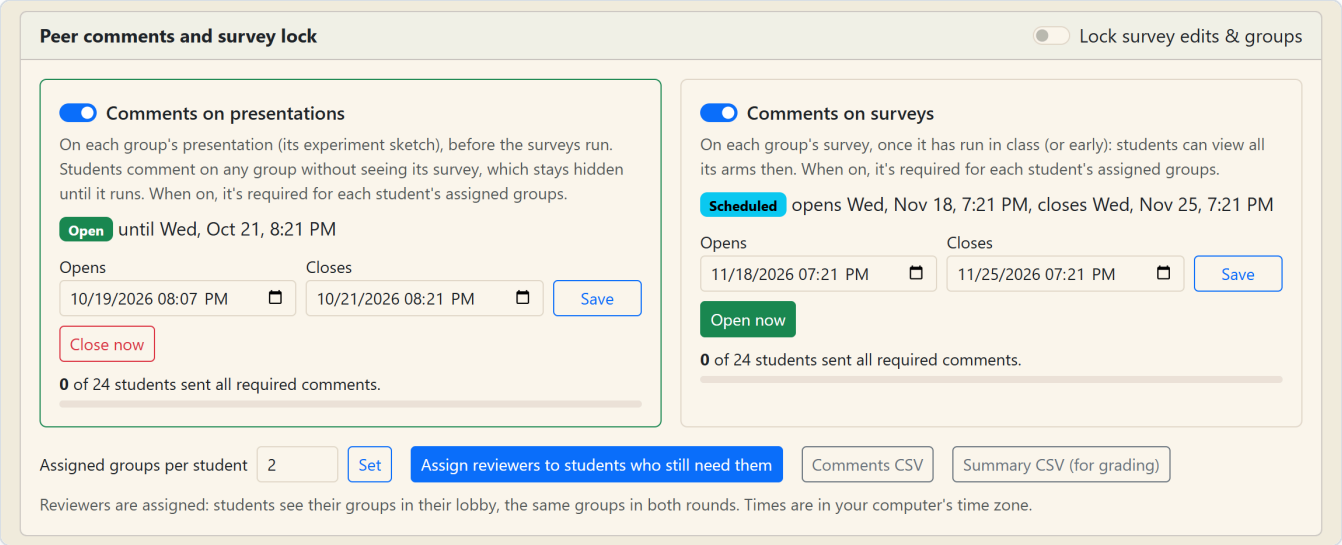



Students not in a group yet.

- **Upload** again any time: new IDs are added, existing students keep their passwords and groups. Optionally remove students who aren't in the file.
- **Add one student** for late additions; **Reset password** if someone forgets theirs (they choose a new one on their next sign-in); **Remove** hides a student who dropped (their past answers stay in your data; you can restore them).
- **Max members per group** and **Max groups per student**: lowering them never removes anyone; full groups just can't take more.

## 4. Peer comments and the survey lock

The **Peer comments and survey lock** card has the lock switch and the two comment rounds.




Presentations open until Wednesday; surveys scheduled for next month.

- **Lock survey edits & groups**: surveys and groups are frozen (you can still edit). Previews and downloads keep working. Lock before the surveys run.
- **Comments on presentations** (off by default): on each group's experiment sketch, before the surveys run. Students comment on any group without seeing its survey, which stays hidden from classmates until it runs, so nobody sees the treatment before answering.
- **Comments on surveys** (on by default): on each group's survey once it has run in class or early; students can view all its arms then.

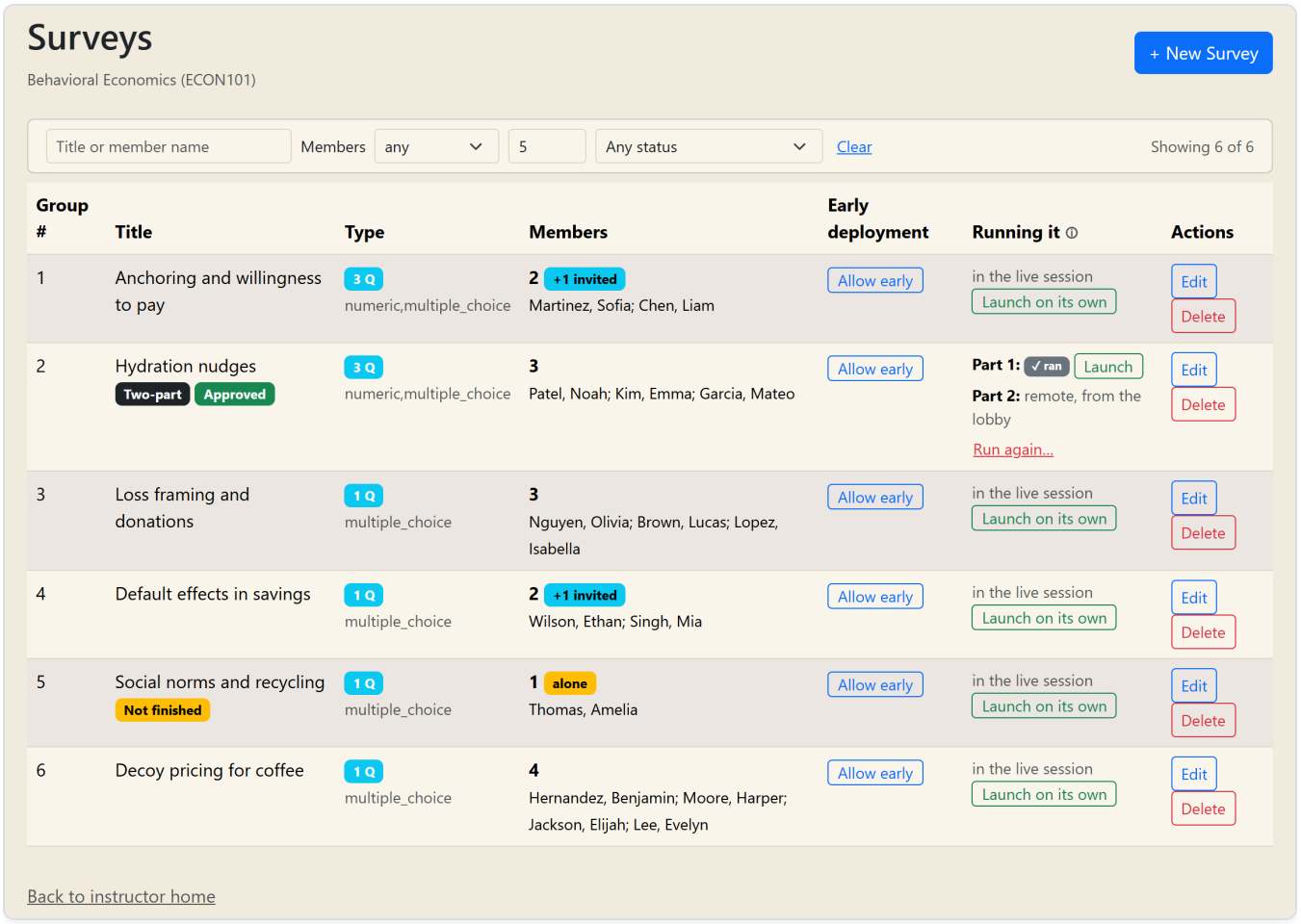
Each round that is on is required for each student's assigned groups (**Assigned groups per student**, same groups in both rounds). For each round:

- **Opens / Closes**: set the dates and times (in your computer's time zone) and **Save**. The round opens and closes on its own; students see the times in their lobby and can save drafts before it opens. Leave **Closes** empty for no end.
- **Open now** / **Close now** for manual control (reopening a closed round removes its end time).
- When a round closes, the groups read its comments, anonymized as "Reviewer 1, 2..." (the same number in both rounds).
- Progress shows how many students sent all their required comments in that round.

**Assign reviewers now** gives every student their groups (never their own; about the same number of reviewers per group). Do it once groups are settled, e.g. before the presentations. Reviewers are also assigned automatically when a round opens or when you lock surveys; clicking again only tops up students who still need groups (e.g. after a merge or a late roster addition). **Comments CSV** and **Summary CSV** have a round column / per-round counts for grading.

## 5. Managing surveys

**Manage surveys** lists every group's survey with its members (count, "alone" and "full" badges, pending invites), status and badges (not finished, two-part, waiting for your approval). Filter by title or member, by number of members, or by status.




- **Edit** opens a group's survey page; you can always edit (saving a survey that has responses deletes them, after a warning).
- **Allow early** lets a one-part survey run before class: the group submits it with an answer-by date, classmates answer from their lobby, and it still runs in class. **Copy email text** gives you an announcement to paste.
- **Running it** shows how each survey (or each part) runs and whether it already ran:
  - **Launch** opens the live dashboard with only that survey or part, e.g. a part that runs at the start of class, or a group that presents on another day.
  - **Next Survey** in the live session skips everything that already ran.
  - **Run again...** (host password) deletes that survey's responses and notification sign-ups so it runs again as new, for a group that changed its survey after running it. Download its data first.

- **View survey** (from a survey's page or the lobby) shows all arms side by side; the flow chart shows each arm's path.

**Question types and rules.** Besides the basic types, groups can use ranking, rating scales and grids, picture choice, budget split, price lists (willingness to pay) and Holt-Laury lottery choices (risk attitude). Per question they can make it optional, shuffle its options, set number limits or mark it as an attention check; per survey, shuffle the question order or add an intro or consent screen. The dashboard draws each type (average ranks, demand curves by arm, share choosing the safe lottery per row...), shows attention-check pass rates, and the downloads add analysis-ready columns (ranks, amounts, switch rows, order shown, attention results). All of these are off by default, so existing surveys are unchanged. The student guide (section 4 and 5) explains them with screenshots.

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## 6. Two-part surveys and notifications

Both are optional; by default a survey has one part, no notifications, and runs in the live session. Groups set them up in two cards at the bottom of their survey page.

### Two-part surveys

Part 2 starts at a question the group chooses. Each part runs **in the live session**, **in class, launched separately** (by you, from Manage surveys), or **remote** (answered from the lobby between the group's dates). Part 2 is only for students who answered part 1, in the same arm; in the live session, others see "This survey is for classmates who answered its part 1" and wait for the next survey. The dashboard marks part-2 questions.

### Notifications

Per arm: a short note respondents read first, then messages, each with a time of day and chosen days. After answering (part 1), respondents add them to their calendar (iPhone and computers: a calendar file; Android: Google Calendar links) or turn on notifications from the site. Times are each phone's local time unless the group picks a time zone.

### Your approval

Notifications and remote dates need your approval: nothing is sent, and remote parts stay closed, until you approve. Review them on the group's **View survey** page, where they're listed under the questions with an **Approve** button. If the group changes them afterwards, they need approval again (the home page lists them).

**Two parts and notifications** ✓ Approved

Part 1: in class, during the live session. Part 2: remote, from the lobby, Wed Oct 7 to Sat Oct 17; everyone who answered part 1 gets a reminder when it opens (10 AM).

**Control**

You'll get a short reminder to stretch every morning this week.

- **Time for a quick stretch!**: daily from Fri Oct 9 to Thu Oct 15 at 9 AM (7 times)

**Nudge**

You'll get a notification about drinking water every day at noon this week, and two evening tips.

- **Time for a glass of water!**: daily from Fri Oct 9 to Thu Oct 15 at 12 PM (7 times)
- **Two liters a day keeps you sharp.**: every 3 days from Sun Oct 11 to Wed Oct 14 at 9 PM (2 times)

Withdraw approval Respondents then get no notifications, and remote parts close.

**Push notifications on the server** need nothing set up: the site creates its own keys on first use and sends due notifications every minute. Optionally set the environment variable `VAPID_SUBJECT` on Render to `mailto:` plus your email, as a contact for the push services. iPhones only get site notifications if the site is added to the Home Screen, so iPhone users are offered the calendar.

## 7. The demo dashboard

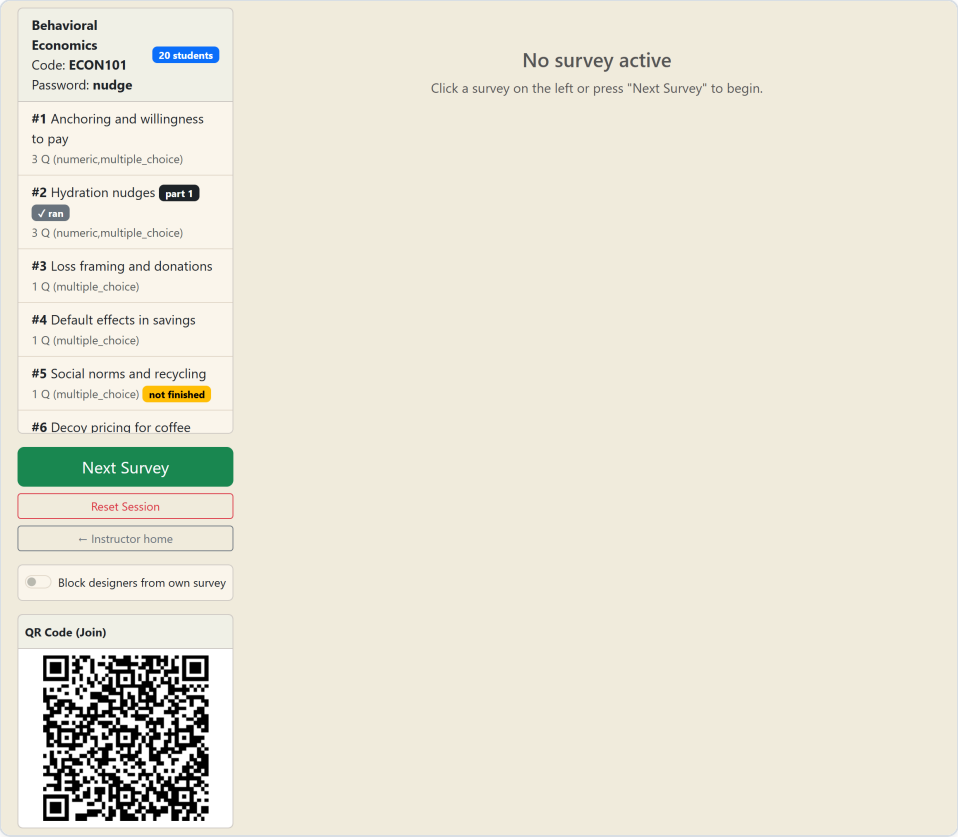
**Demo dashboard** (Surveys card on the home page) lets you click through every survey's questions, images and charts with made-up answers, without going live. Use it to check the surveys before class or to walk the class through the designs.




| Arm | N | Mean | Median | Std Dev | Min | Max |
|---|---|---|---|---|---|---|
| **Control** | 23 | 58.35 | 58 | 14 | 37 | 101 |
| **High anchor** | 17 | 37.53 | 38 | 10.5 | 18 | 60 |

## 8. Class day: the live session

1. On the home page, click **Start live session** and project it. Students join from their lobby, or by scanning the QR code with the classroom code and password.
2. Click a survey on the left, or **Next Survey**, to start it. Students' phones show it within a few seconds; each student always gets the same arm.
3. Click **Refresh Results** for the latest charts (they don't update on their own, to keep the site fast with a full class).
4. **Next Survey** moves on in group order, skipping surveys or parts that already ran; **Reset Session** stops the current survey (nothing is deleted).




- Two-part surveys show which part runs here ("part 2"); a "✓ ran" badge marks what already ran (click it to show its results again).
- **Block designers from own survey** shows group members "You designed this survey" instead of their own questions.
- Surveys run on another platform are listed for the order but don't start here.

## 9. Peer feedback

See section 4 for the rounds and their schedules. Comments are drafts until sent; students can jot private notes right after answering a live survey, which become drafts in the survey round. Groups read the comments as "Reviewer 1, Reviewer 2" once each round closes; you see names. **Feedback comments** and **Feedback summary** (per round) are in the Data card and the peer comments card, for grading.

## 10. Data and downloads

**Download everything (.zip)** on the home page has every file plus the uploaded images:

| File | Contents |
|---|---|
| all_responses.csv | Every answer, with names, IDs, the respondent ID, arm, time on the question, left-page and timeout flags, and part. |
| surveys_config.csv | What each arm showed: question texts and options. |
| participants.csv | The roster key: names, IDs and the random respondent IDs students see. |
| survey_designers.csv / survey_participation.csv | Who designed each survey; who answered each one. |
| feedback.csv / feedback_summary.csv | Comments and required-comment counts, for grading. |
| reminder_plans.csv | Each group's notifications per arm, how its parts run, and approval. |
| reminder_actions.csv / reminder_notifications.csv | Who downloaded the calendar file, opened Google Calendar links, turned notifications on; each notification sent, delivered and tapped. |

Groups download their own survey's responses, anonymized (random respondent IDs only), in wide (one row per respondent) or long (one row per answer) format.

## 11. Deploying and backups

- The site runs on Render with a persistent disk ( `/data` ) for the database and images. Pushing to the repository deploys it.
- Before deploying, download the zip, and if possible a copy of `/data/app.db`. Don't deploy during class: the site restarts.
- Updates only add to the database; existing classrooms, students, groups and surveys are kept.

## 12. Troubleshooting

| Problem | What to do |
|---|---|
| A student can't sign in | Check they're on the roster (not removed); reset their password if needed. |
| A survey doesn't appear on phones | Students poll every 3 seconds; have them go back to the lobby and click Join. Check the survey isn't a part that's remote, or that it isn't already "ran" (click it to show it). |
| A group needs to run its survey again | Manage surveys → Run again… (download its data first). |
| A remote part doesn't open | It needs your approval and its dates; see "to approve" on the home page. |
| "0 survey answers collected" | Normal before surveys run; it counts answers, not surveys built. |